THE SEPARABLE DEGREE EQUALS THE NUMBER OF EMBEDDINGS INTO AN ALGEBRAIC CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The unique extension claim ('by the result of the previous paragraph') is the just-proved purely inseparable case applied to K/K_s: there is exactly one K-embedding of K into F-bar over a given embedding of K_s.

Editorial note. The label lemma-separable-degree names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 1943-1954: the definition of the separable degree as the degree of the separable subextension
- Same chapter: lemma-finite-purely-inseparable (the x^p - alpha_i^p tower for purely inseparable extensions), lemma-count-embeddings-explicitly (|Mor_F(K, F-bar)| = product deg_s(P_i)), lemma-separable-first (the separable subextension K_s), lemma-separable-equality (|Mor_F(K_s, F-bar)| = [K_s : F] with equality iff separable)

Source reference. Original reference 09H8: sources/stacks-fields/pinned-original-source.tex, source lines 1543–1543. Prerequisite pointer invoked at 1979; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09HI: sources/stacks-fields/pinned-original-source.tex, source lines 1886–1886. Prerequisite pointer invoked at 1977; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09HA: sources/stacks-fields/pinned-original-source.tex, source lines 1601–1601. Prerequisite pointer invoked at 1986; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 030K: sources/stacks-fields/pinned-original-source.tex, source lines 1921–1921. Prerequisite pointer invoked at 1985; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Definition 0.1. Let E/F be an algebraic field extension. Let E_{sep} be the subextension found in Lemma 030K.

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) The integer $[E_{sep} : F]$ is called the *separable degree* of the extension. Notation $[E : F]_s$.
(2) The integer $[E : E_{sep}]$ is called the *inseparable degree*, or the *degree of inseparability* of the extension. Notation $[E : F]_i$.

Lemma 0.2. *Let K/F be a finite extension. Let $\overline{F}$ be an algebraic closure of F. Then $[K : F]_s = |\operatorname{Mor}_F(K, \overline{F})|$.*

Proof. We first prove this when K/F is purely inseparable. Namely, we claim that in this case there is a unique map $K \to \overline{F}$. This can be seen by choosing a sequence of elements $\alpha_1, \ldots, \alpha_n \in K$ as in Lemma 09HI. The irreducible polynomial of α_i over $F(\alpha_1, \ldots, \alpha_{i-1})$ is $x^p - \alpha_i^p$. Applying Lemma 09H8 we see that $|\operatorname{Mor}_F(K, \overline{F})| = 1$. On the other hand, $[K : F]_s = 1$ in this case hence the equality holds.

Let's return to a general finite extension K/F. In this case choose $F \subset K_s \subset K$ as in Lemma 030K. By Lemma 09HA we have $|\operatorname{Mor}_F(K_s, \overline{F})| = [K_s : F] = [K : F]_s$. On the other hand, every field map $\sigma' : K_s \to \overline{F}$ extends to a unique field map $\sigma : K \to \overline{F}$ by the result of the previous paragraph. In other words $|\operatorname{Mor}_F(K, \overline{F})| = |\operatorname{Mor}_F(K_s, \overline{F})|$ and the proof is done. □